

Full length article

Enhanced energy storage performance of 0.85BaTiO₃–0.15Bi(Mg_{0.5}Hf_{0.5})O₃ films via synergistic effect of defect dipole and oxygen vacancy engineering

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ABSTRACT

Dielectric capacitors are widely used in electronic devices due to their ultra-fast charge/discharge rate and ultra-high power density, but their lower energy density and poor thermal stability limit their further application. In contrast to the traditional strategy of suppressing defects, this work combines oxygen vacancies with defect dipoles to improve the breakdown strength and polarization behavior of ferroelectric films. Low concentration of oxygen vacancies and defect dipoles can trap charge carriers and increase breakdown strength, but if the concentration is too high, it can easily make films prone to breakdown. Moreover, the defect dipoles can reduce P_r by providing intrinsic restoring force for polarization switching, while excessive defect dipoles and oxygen vacancies can pin domain walls and increase P_r . By delicately controlling the concentration of oxygen vacancies and defect dipoles in the film, the BT-BMH film deposited at 0.135 mbar achieved the maximum breakdown strength and slim P - E loops, inducing the energy density to reach 108.9 J·cm⁻³ with an efficiency of 79.6 % at room temperature and excellent thermal stability in the wide temperature range of -100~350 °C with the energy density of 69.1 J·cm⁻³. This work reveals the important significance of reasonable defect control for improving energy storage performance and provides an effective method for developing high-performance dielectric capacitors.

1. Introduction

Dielectric capacitors, unlike batteries or electrochemical capacitors, without the chemical reactions in the energy storage process, have the merits of high power density, ultra-fast charge and discharge rate and stable performance, posing a huge potential employed in pulsed power systems and aerospace, and so on [1–3]. Unfortunately, the relatively low energy density of dielectric capacitors severely limits their further widespread application. Continuous efforts have been made by researchers to increase energy density, especially Pb-based materials with high maximum polarization caused by unique lone pair electronic 6s² configuration, which have previously been considered excellent candidates for dielectric energy storage. However, with environmental regulations prohibiting the use of lead, the development of Pb-free dielectric capacitors with high energy storage properties is of great practical importance [4].

Generally, the recoverable energy density for dielectric films can be

calculated from the P - E loops based on $W_{re} = \int_{P_r}^{P_m} E dP$, where E is the applied electric field strength, P_m and P_r are the maximum polarization and the remnant polarization, respectively. The energy efficiency is defined as $\eta = \frac{W_{re}}{W_{stored}} \times 100\%$, where the stored energy density is $W_{stored} = \int_0^{P_m} E dP$. From the above formula, it can be concluded that dielectric films with high P_m/P_r and high breakdown strength (E_b) are vital for the overall enhancement of energy storage performance.

Among various dielectric materials, relaxor ferroelectrics are able to rapidly respond to the electric field due to their disordered polar structure (polar nanodomains (PNDs)/polar nanoregions (PNRs)) with high dynamic properties, which are beneficial for the small P_r with slim P - E loops, and favoring for excellent energy storage properties [2,5]. Thus, extensive researches have been carried out on relaxor ferroelectrics to improve the energy storage performance. The interface engineering in multilayer systems (Ba_{0.7}Ca_{0.3}TiO₃-BaZr_{0.2}Ti_{0.8}O₃) can effectively inhibit the spread of current trees and increase breakdown strength [6]. The high-entropy strategy introduces amorphous-like

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phase and lattice distorted nano-crystalline grains in $\text{Bi}_{3.25}\text{La}_{0.75}\text{Ti}_{3-3x}\text{Hf}_x\text{Zr}_x\text{Sn}_x\text{O}_{12}$ films, which help to reduce the polarization switching hysteresis and enhance the E_b [7]. Growth orientation modulation of $0.85\text{BaTiO}_3\text{--}0.15\text{Bi}(\text{Mg}_{0.5}\text{Zr}_{0.5})\text{O}_3$ relaxor ferroelectric films can alter the strain anisotropy of films, making it easier to form weakly coupled nanodomains with low flip energy barriers, exhibiting large maximum polarization and small hysteresis loss [8]. The combination of the amorphous matrix with high breakdown strength and the crystalline material with high maximum polarization contributes to the overall energy storage performance in the $\text{Bi}_3\text{NdTi}_4\text{O}_{12}$ films [9]. For $\text{BiFeO}_3\text{--}\text{BaTiO}_3\text{--}\text{SrTiO}_3$ systems, the polymorphic nanodomain design realize the coexistence of tetragonal and rhombohedral nanodomains in a cubic matrix, which make it maintain high polarization and achieve high energy density [10]. Unfortunately, limited by the negative correlation between breakdown strength and maximum polarization, the enhancement of the energy storage performance of relaxor ferroelectric films is usually at the sacrifice of maximum polarization, which severely hinders the further improvement of the energy storage performance [11,12]. It is difficult to overcome this inherent limitation using methods such as solid solution alone, so how to break this inherent limitation by adopting appropriate methods is still an issue worth exploring.

Recent studies have shown that defect engineering appear to offer a feasible method to break the inverse relationship. Normally, low concentrations of oxygen vacancies act as trap-filling centers to capture charge carriers, but high concentrations of oxygen vacancies form electron transport paths, exacerbating leakage, degrading the energy storage performance of the capacitor, and interacting with the ferroelectric domain walls to degrade the polarization characteristics [13]. Pan et al. introduced SrTiO_3 into the BiFeO_3 system to inhibit the valence change of Fe^{3+} while reducing the generation of oxygen vacancies, which greatly enhanced the electrical insulation and breakdown strength of the film, with a breakdown strength of up to $4.46 \text{ MV}\cdot\text{cm}^{-1}$, twice that of pure BiFeO_3 , and achieved a high energy density of $70.0 \text{ J}\cdot\text{cm}^{-3}$ [14]. In addition to reducing the concentration of oxygen vacancies by methods such as doping or solid solution, cationic vacancies can be used to construct defect dipoles with oxygen vacancies to enhance energy storage performance. Li et al. introduced Ti-O defect complex into Mn-doped SrTiO_3 system, resulting in local symmetry breaking and local lattice strains, to form ultrafine polar nanoclusters with considerable dielectric polarization and negligible polarization hysteresis, achieving superior comprehensive performance [15]. Similarly, Pan et al. introduced oxygen vacancies and bismuth vacancies into $30\text{BiFeO}_3\text{--}35\text{BaTiO}_3\text{--}35\text{SrTiO}_3$ films to create deep energy level defect complexes with high carrier activation energies, which significantly improved the resistivity of the films and increased the E_b by 150 % [16]. Although research on enhancing energy storage performance via defect engineering has been carried out, how to modulate the defect dipoles and oxygen vacancies by oxygen pressure to affect the energy storage properties and breakdown behavior of dielectric films is still an issue worth investigating.

$\text{BaTiO}_3\text{--}\text{Bi}(\text{M})\text{O}_3$ (BT-BiMO₃, M represents a single trivalent cation or two cations with an average valence state of trivalent) is a novel weakly-coupled relaxor ferroelectric material with excellent dielectric properties and good stability under high temperature and strong electric fields [17]. The single pair electronic in the $6s^2$ orbital of Bi^{3+} is able to hybridize with the $2p$ orbital of O^{2-} , which facilitates the maximum polarization and compensates for the insufficient A-site polarization contribution of BaTiO_3 [18]. In addition, the differences in ionic radius and valence state of the BT-BiMO₃ can lead to a high degree of disorder in the local structure and promote the formation of polar nanodomains, which is beneficial for improving the relaxation properties and forming slim $P\text{--}E$ loops [19]. Finally, MgO is a component with high insulating properties, and HfO_2 has a large band gap (5.7 eV), both of which are favorable for increasing the breakdown strength [20,21]. Thus, in this work, the relaxor ferroelectric $0.85\text{BaTiO}_3\text{--}0.15\text{Bi}(\text{Mg}_{0.5}\text{Hf}_{0.5})\text{O}_3$ (BT-BMH) with high P_m and E_b was chosen as the basic material, and the

effects of defect dipoles and oxygen vacancies on the energy storage properties of the films were systematically investigated by varying the oxygen pressure. We introduced different concentrations of Bi-O defect dipoles and oxygen vacancies into BT-BMH films by controlling the oxygen pressure during film growth process, and the results showed that a moderate concentration of defect dipoles and oxygen vacancies is beneficial for increase the resistivity and improve the breakdown strength. Ultimately, the BT-BMH films grown at 0.135 mbar achieved the highest energy density ($\sim 108.9 \text{ J}\cdot\text{cm}^{-3}$) and efficiency ($\sim 79.6 \%$) with excellent thermal stability in the wide temperature range ($-100\sim 350^\circ\text{C}$) under the synergistic effect of defect dipole and oxygen vacancy.

2. Experiment section

2.1. Fabrication of epitaxial films

BT-BMH ceramic target was prepared by conventional solid-state reaction. 10 mol% Bi was added in excess to compensate for the Bi volatilization during target sintering and film fabrication. All the films with a thickness of $\sim 250 \text{ nm}$ were deposited on (001)-oriented 0.7 wt% Nb-doped SrTiO_3 (Nb:STO) single-crystal substrates using a radio frequency magnetron sputtering system, and the substrate temperature was maintained at 700°C during deposition. The BT-BMH films were grown at different total Ar/O_2 mixture pressure of 0.100 mbar, 0.135 mbar and 0.170 mbar with the mixed ratio of 1:1, and all of them were in-situ annealed for 15 min under the pressure of 200 mbar after finishing the deposition, then naturally cooled down to room temperature. In order to measure the dielectric and ferroelectric properties, Pt top electrodes (square electrodes with dimensions of $200 \mu\text{m} \times 200 \mu\text{m}$) were deposited through a stainless-steel shadow mask using a sputtering system.

2.2. Characterization

The phase structures and epitaxial properties of the BT-BMH films were characterized by a high-resolution X-ray diffraction (HRXRD, PANalytical X'Pert MRD) with the X-ray of $\text{Cu K}\alpha$ ($\lambda=1.540598 \text{ \AA}$). The scanning transmission electron microscopy (STEM) imaging and energy dispersive spectrometer (EDS) mapping were performed on the scanning transmission electron microscope (JEOL ARM 200F). The chemical composition and states were analyzed by X-ray photoelectron spectroscopy (XPS, ESCALAB Xi+). The dielectric constant (ϵ_r) and dielectric loss ($\tan\delta$) as a function of frequency (1 kHz–1 MHz) of films were conducted by the Keysight E4980AL Precision LCR Meter with a perturbation voltage of 1 V. The electric breakdown value (E_b) and the $P\text{--}E$ loops under 1 kHz and the first-order reversal curve (FORC) loops under 40 Hz were conducted by using TF2000 Analyzer. The leakage current density curve of the BT-BMH films was measured by using a TF2000 Analyzer with a step scan duration of 2 s. The fast discharge experiments were performed by a capacitor discharge test system (PolyK Technologies PK-CPR 1701).

3. Results and discussion

The $\theta\text{--}2\theta$ scans (Fig. 1a) of the films show only (00 l) Bragg reflections, indicating that all of the films are high-quality single-crystal c-oriented perovskite structures with no secondary phases. Magnified diffraction peaks of the BT-BMH films at approximately $40^\circ\text{--}50^\circ$ are shown in Fig. 1b. It is evident that as the oxygen pressure increases, the (002) diffraction peak gradually shifts towards a lower angle, indicating the increase of out-of-plane lattice c of the BT-BMH film. The observed lattice c expansion could potentially be attributed to variations of defect, such as ionic vacancies [16]. The epitaxial relationship between the BT-BMH film and the Nb:STO substrate is confirmed to be $(001)_{\text{BT-BMH}} \parallel (001)_{\text{Nb:STO}}$ and $[100]_{\text{BT-BMH}} \parallel [100]_{\text{Nb:STO}}$ by φ scan measurements of

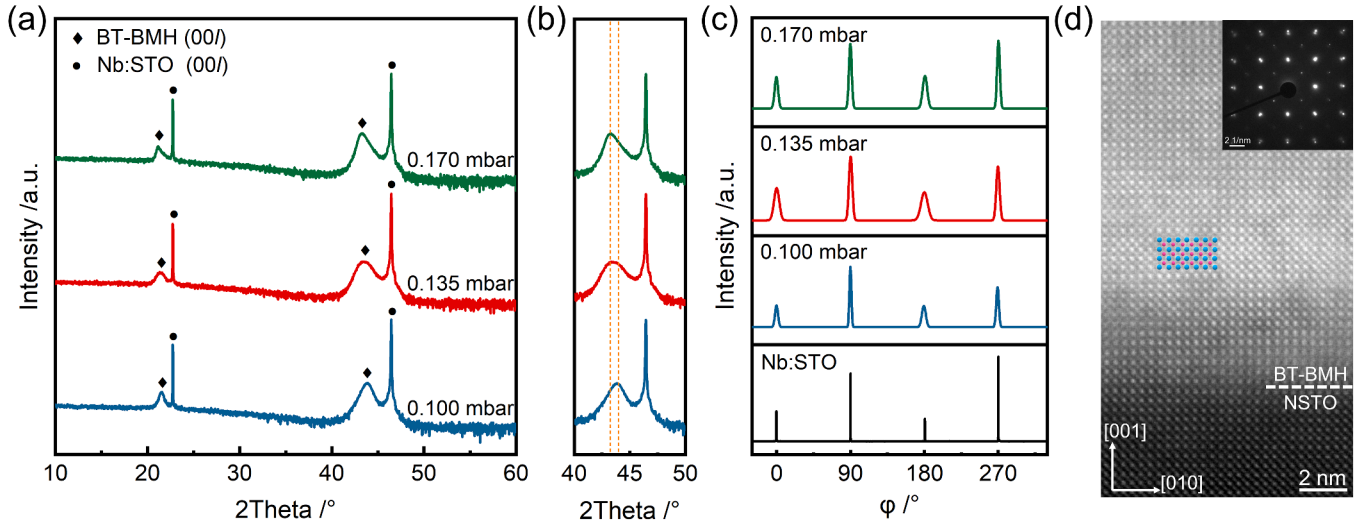


Fig. 1. (a) Typical XRD θ - 2θ scanning patterns of the BT-BMH films. (b) Enlargements of the (002) peaks. (c) ϕ -Scans taken around the {101} reflection of the BT-BMH films and Nb:STO substrate. (d) HAADF-STEM image of the BT-BMH film grown at 0.135 mbar; the inset is the corresponding SAED pattern at the BT-BMH/Nb:STO interface.

the {101} crystal plane (Fig. 1c). The STEM (Fig. 1d) and SAED image (inset of Fig. 1d) further illustrate that the films have a good epitaxial structure.

To investigate the differences in the internal defects of the films grown under different oxygen pressures, the element valences and contents were analyzed by XPS. The C 1s peak at the core level (284.8 eV) was used as a reference for calibrating the binding energy values. Fig. 2a shows the O 1s core-level XPS spectra of the films, which consist of three absorption peaks, where the 529.1 eV peaks correspond to the vibrational absorption peak of lattice oxygen (O_L), the peaks at 531 eV are attributed to the presence of oxygen vacancies (O_V) in the film, and the absorbed oxygen or hydroxyl groups on the film surface are represented by the peaks at 532.38 eV [22]. Usually, the change in oxygen vacancy concentration is linked with the change in the ratio of oxygen vacancy to lattice oxygen content (O_V/O_L) [22]. As shown in Fig. 2d, the O_V/O_L value gradually increases as the oxygen pressure rises, indicating that the content of oxygen vacancy also increases. Generally speaking, there should be a negative correlation between the oxygen pressure and the content of oxygen vacancies. However, if cation vacancies and oxygen vacancies form defect dipoles, the concentration of oxygen vacancies will decrease [23]. Here, we infer that there are probably defect dipoles in the film, which cause the oxygen vacancy content to increase as the oxygen pressure increases.

There are two sources of cation vacancies in the film: the valence changing of the Ti element ($Ti_{Ti}^{\prime}$) and the volatilization of the Bi element ($V_{Bi}^{\prime}$). As shown in Fig. 2b, the Ti 2p bimodal peaks in all films are composed of Ti 2p_{3/2} (457.92 eV) and Ti 2p_{1/2} (463.66 eV), corresponding to stable Ti^{4+} states, indicating that it is impossible to have $Ti_{Ti}^{\prime}$ cation vacancies [24]. The Bi 4f doublet consists of Bi 4f_{7/2} (158.94 eV) and Bi 4f_{5/2} (164.25 eV), corresponding to the Bi^{3+} state, as shown in Fig. 2c [25]. The content of Bi element is shown in Fig. 2d, and it can be seen that the Bi content rises with increasing oxygen pressure, which suggests that the volatilization of Bi elemental is more severe at low oxygen pressure, resulting in higher concentrations of Bi vacancies. As we known, the Bi vacancies is prone to interact with oxygen vacancies and form Bi-O defect dipoles ($V_{Bi}^{\prime} - V_O^{\prime}$). The higher Bi vacancies means higher defect dipoles at 0.100 mbar, and induce the concentration of oxygen vacancies becomes lower (Fig. 2d), while the film grown at 0.170 mbar with low Bi vacancies contributions to the higher oxygen vacancies (Fig. 2d) and leads to the lower defect dipoles. The film grown at 0.135 mbar has a moderate concentration of oxygen vacancies and defect dipoles. This phenomenon also can be derived from the change of

the P - E loops. For a normal relaxor ferroelectric material, the P - E loops should be symmetrical about the origin, whereas as shown in Fig. 2e, different degrees of deviation from the origin are observed in the P - E loops of BT-BMH films grown under different oxygen pressures, which is a typical characteristic of the presence of defect dipoles [26,27]. The inset of Fig. 2e shows the degree of deviation of the center of the P - E loop deviates from the origin at different oxygen pressures. It can be seen that the shift decreases with increasing oxygen pressure, suggesting that the defect dipole effect and the concentration of Bi-O defect dipole degrades as the oxygen pressure increases [27,28].

Oxygen pressure not only has a significant effect on the internal defect structure of the film but also has an impact on its sub-grain structure. RSM was used to explore the structural variation induced by different deposited oxygen pressures, as shown in Fig. 3a-c. It can be seen that the symmetric RSM maps of the films grown under different oxygen pressures exhibit broadening in the Q_x direction, indicating a certain degree of tilting and twisting between different sub-grains in the films [29]. The tilt and twist angles between sub-grains and the sub-grain size can be quantified by fitting the RC curves using Lee mode [30]:

$$FWHM = \sqrt{(\Gamma_{tilt} \cos \chi)^2 + (\Gamma_{twist} \sin \chi)^2 + (2\pi/L)^2} / K_{hkl}^2$$

where FWHM is the full width at half maximum of the diffraction plane, Γ_{tilt} and Γ_{twist} are the out-of-plane tilt and in-plane twist angles of the sub-grains, respectively. χ is the angle between the (hkl) plane and the film surface, L is the coherence length of the sub-grains, including vertical grain size L_v and lateral grain size L_s , $K_{hkl} = 4\pi \sin \theta / \lambda$ is the scattering vector of the (hkl) crystal plane, where θ is the Bragg diffraction angle and λ is the diffraction wavelength. As seen in Fig. 3d-e, the out-of-plane tilt angles of the sub-grains in the films grown at 0.100 mbar, 0.135 mbar and 0.170 mbar were calculated to be 2.42°, 2.58° and 1.37°, respectively, and the in-plane twist angles were 2.10°, 2.78° and 1.94°, respectively, by fitting the FWHM of the (001), (002), (101) and (202) crystal planes of the BT-BMH films (Fig. S1). The 0.135 mbar film has the largest out-of-plane tilt angle and in-plane twist angle, which facilitates the refinement of the P - E loops and increases the breakdown strength, thus contributing to the energy storage performance [29].

In addition, as shown in Fig. 3d and e, it can be observed that both the lateral sub-grain size L_s and coherence length L decrease gradually with increasing oxygen pressure. In general, changing the oxygen pressure can affect the diffusion rate and deposition rate of atoms on the

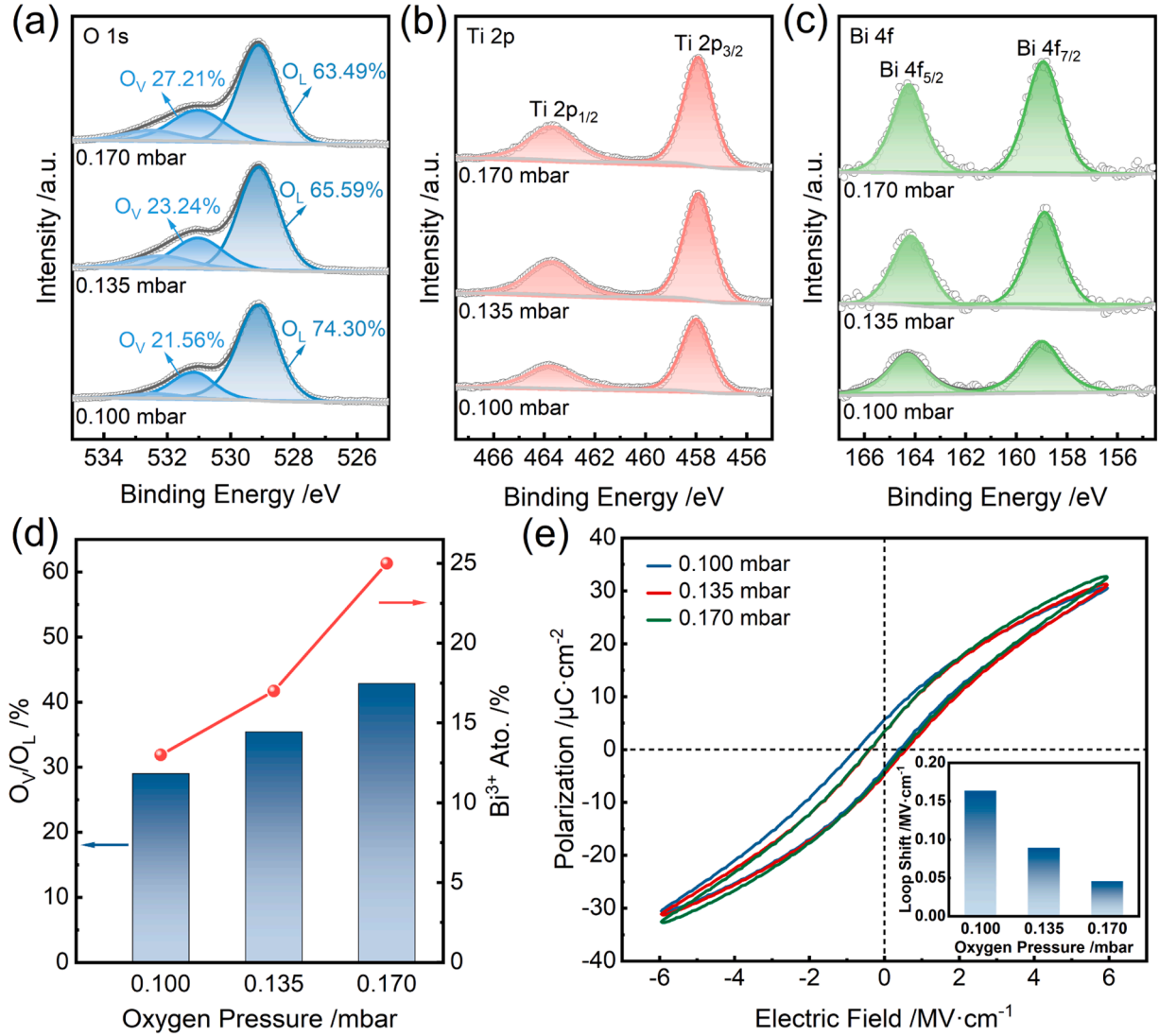


Fig. 2. XPS spectra of O 1s (a), Ti 2p (b), Bi 4f (c) peak of BT-BMH films deposited under different oxygen pressures. (d) Variation of the O_V/O_L ratio and the Bi^{3+} fraction with the change of oxygen pressure. (e) Bipolar P - E loops of BT-BMH films under different oxygen pressure. The inset shows the loop shift of the P - E loops as a function of oxygen pressure.

substrate surface, thereby affecting thin film growth and grain size. The probability of collisions between sputtered atoms and gas molecules is lower and the atoms arrive at the substrate with higher kinetic energy at low oxygen pressures, which promotes atom diffusion on the substrate surface and contributes to the formation of larger grains [31]. High oxygen pressure causes more atomic deposition per unit area and promotes nucleation, which is beneficial for grain size reduction [32,33]. Therefore, as the oxygen pressure increases, the diffusion rate of sputtered atoms decreases while the deposition rate increases, resulting in the atoms deposited on the substrate surface under high oxygen pressure pile up and decrease grain size. This also can be confirmed from the increased wave surface, as shown in the cross-section STEM dark field images of BT-BMH films grown at different oxygen pressures (Fig. 4). The insets of Fig. 4 show the corresponding selected area electron diffraction (SAED) patterns of the films, all of which exhibit clear single crystal diffraction peaks accompanied by varying degrees of stretching. The strongest stretching of the diffraction spots on the 0.135 mbar film indicates a larger angle of tilt and twist between its sub-grains, further verifying the FWHM fitting results.

The microstructure definitely influences the breakdown strength, which is a key parameter in determining the energy density of dielectric capacitors. As shown in Fig. 5a, the statistical breakdown strength E_b

derived from the two-parameter Weibull distribution analysis is $8.06\ MV \cdot cm^{-1}$, $8.60\ MV \cdot cm^{-1}$ and $7.42\ MV \cdot cm^{-1}$ for the film grown at 0.100 mbar, 0.135 mbar and 0.170 mbar, respectively. Obviously, the 0.135 mbar achieved the highest breakdown strength, which is critical to achieving high energy density, and we attribute the improvement in breakdown strength to three factors. Firstly, oxygen vacancy can act as both the trap center and carrier transport path, which is dependent on the concentration of the oxygen vacancy [34,35]. Meanwhile, a suitable concentration of Bi-O defect dipoles can interact with free charges to generate deep-level defect states, reducing the activation of charge carriers and increasing the resistivity [13,16,31]. The moderate concentration of defect dipoles and oxygen vacancies in the film grown at 0.135 mbar make it exhibit the largest breakdown strength. The film grown at 0.100 mbar, which have an excessively high concentration of defect dipoles, tend to form an enhanced local electric field, making the film susceptible to breakdown [36]. While the film grown at 0.170 mbar are more susceptible to leakage due to an excessive concentration of oxygen vacancies acting as carrier transport pathways [37]. Secondly, the large out-of-plane tilt angle facilitates the formation of weakly coupled nanodomains, which can refine the hysteresis loops and lead to polarization delay, thereby contributing to the enhancement of the breakdown strength [29]. Thirdly, the interfaces induced by the twisting

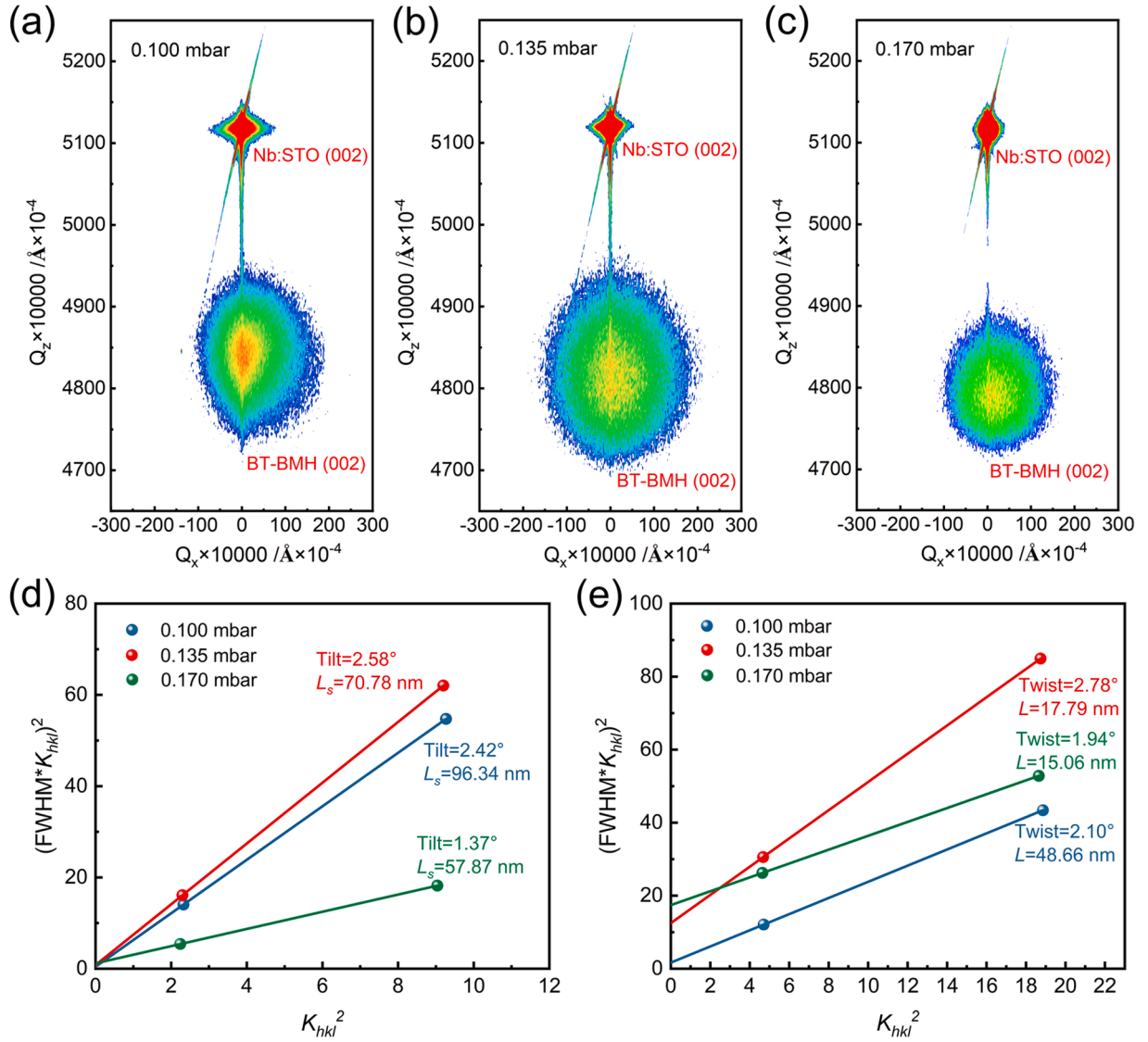


Fig. 3. (a)-(c) Reciprocal space mappings (RSM) taken around the (002) reflection for the films grown at different oxygen pressures. (d) The results of fitting the FWHM of (001) and (002) rocking curve based on model of Lee. (e) The results of fitting the FWHM of (101), (202) rocking curve based on model of Lee.

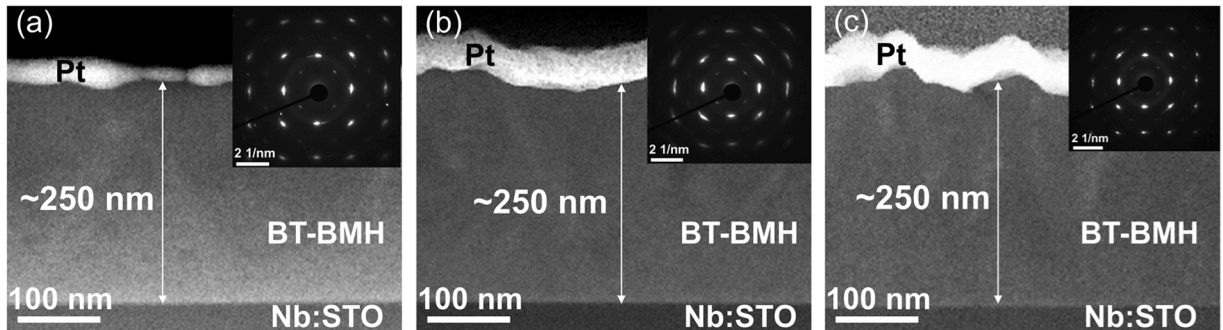


Fig. 4. (a), (b) and (c) Low magnification dark field STEM images of BT-BMH films grown on (001)-oriented Nb-doped SrTiO₃ (Nb:STO) substrates at 0.100 mbar, 0.135 mbar, 0.170 mbar, respectively. Insets of the figures show the corresponding selected area electron diffraction (SAED) patterns of BT-BMH films.

of the sub-grains make the electrons left and right bound, which can reduce the electron transport speed and improve the breakdown strength. Therefore, the film grown at 0.135 mbar with the largest tilt angle and twist angle as well as a suitable defect structure exhibits the maximum breakdown strength.

To further reveal the relationship between breakdown strength and defect structures, the leakage currents were shown and fitted in Fig. 5b-e and the fitting results are summarized in Fig. 5f. Under a low electric field, all thin films undergo a gradient transition from Ohmic conduction mechanisms to Trap-Filled-Limited (TFL) conduction mechanisms and

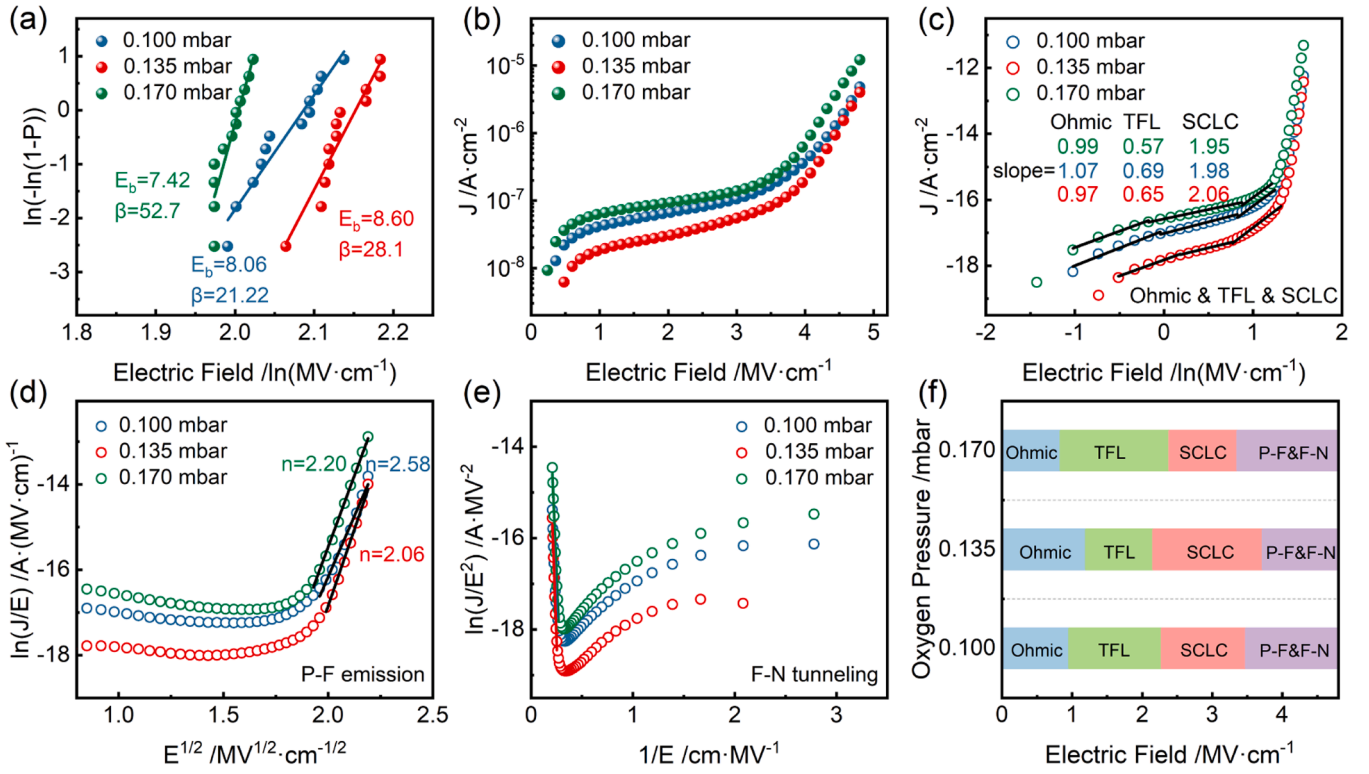


Fig. 5. (a) Weibull distribution for the dielectric breakdown strength. (b) The leakage current density of BT-BMH films as a function of biased DC electric field. Leakage current density fitted by conduction mechanism (c) Ohmic conduction, trap filled limited (TFL) and space charge limited conduction (SCLC), (d) Poole-Frenkel (P-F) emission and (e) Fowler-Nordheim (FN) tunneling. (f) Summary of conduction mechanism in BT-BMH films as a function of electric field.

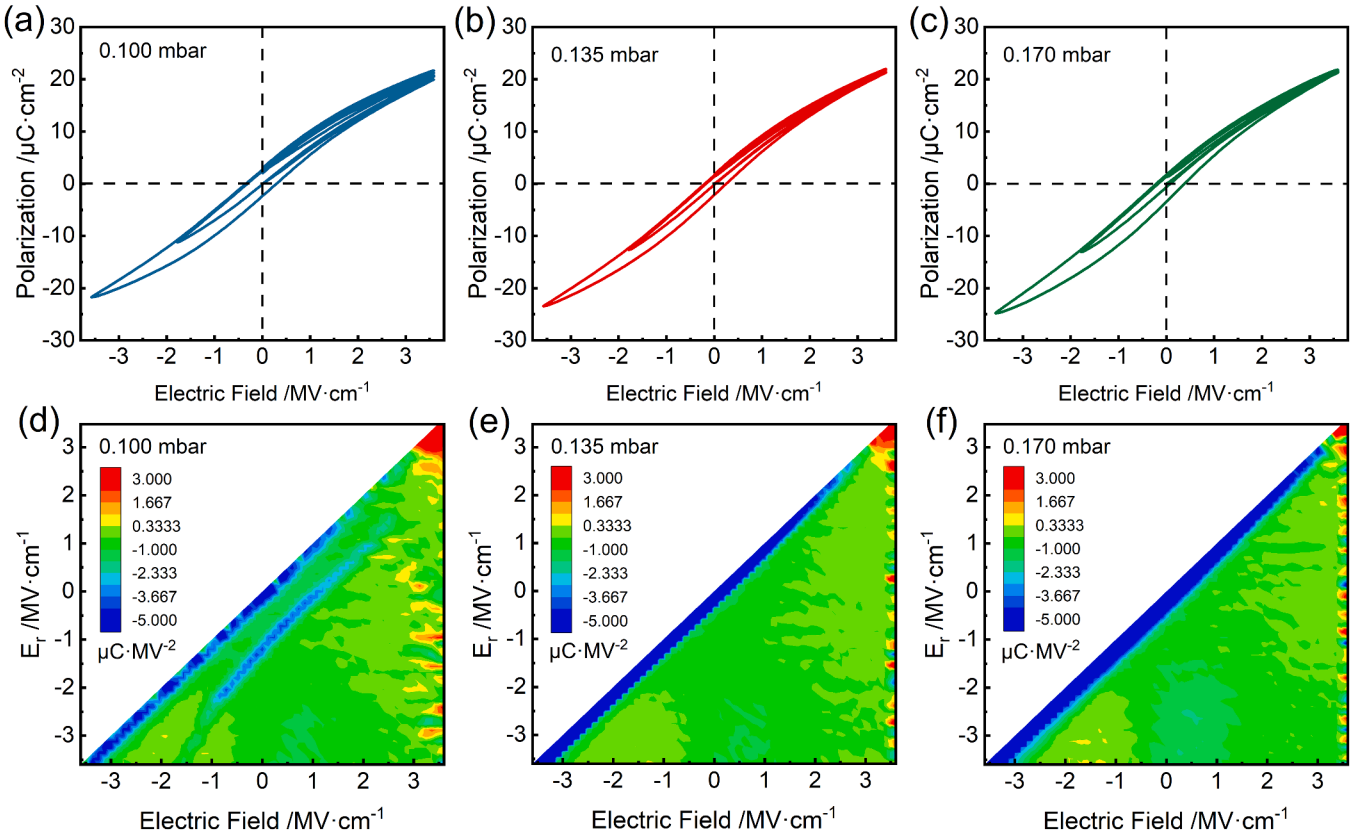


Fig. 6. The FORC loops of (a) 0.100 mbar, (b) 0.135 mbar, (c) 0.170 mbar. (Only 4 out of the total 60 loops are shown for clarity.) (d)-(f) FORC distribution $\rho(E_r, E_f)$ of the BT-BMH films under different oxygen pressures.

finally enter into the space charge limited conduction (SCLC) mechanisms as the electric field increases [38,39]. From Fig. 5f, it can be seen that the ohmic and SCLC conduction mechanisms occur over a wider electric field range in the film grown at 0.135 mbar, which can delay the charge carriers to enter into emission and tunneling. When the electric field strength is further increased to high fields, the conduction mechanism converts to Poole-Frenkel (P-F) emission and Fowler-Nordheim (F-N) tunneling [40]. The threshold field (E_{th}), defined as the transition electric field from the SCLC mechanism into the P-F emission mechanism, is 3.48 MV·cm⁻¹, 3.72 MV·cm⁻¹ and 3.36 MV·cm⁻¹ for 0.100 mbar, 0.135 mbar, and 0.170 mbar films, respectively. The relatively low threshold field for the film grown at 0.170 mbar and 0.100 mbar is due to highest oxygen vacancy concentration and defect dipole concentration in these two films, respectively, since the P-F emission is a volume limited conduction mechanism and the activation of defect carriers is the main source [40]. This result suggests that a suitable defect structure can effectively trap carriers and delay their emission into the conduction band to participate in the conduction process, thereby reducing the leakage current density of the film and effectively increasing breakdown strength of the film grown at 0.135 mbar.

In addition to the breakdown strength, the polarization behavior of dielectric films is another key parameter in determining energy storage performance. The first order reversal curve (FORC) is a helpful method used to provide a reasonable description of the polarization characteristics. Here, E_{max} is set to be 3.6 MV·cm⁻¹, $\Delta E = \Delta E_r = 0.12$ MV·cm⁻¹, and a total of 60 FORC loops were measured for calculation (Fig. 6a-c). The FORC density distribution is calculated by using the formula $\rho(E_r, E) = \frac{1}{2} \frac{\partial^2 (E_r, E)}{\partial E_r \partial E}$, where $\rho(E_r, E)$, E and E_r are the distribution density, the real electric field and the reversal electric field, respectively. The FORC density distributions of the films grown at different oxygen pressures are shown in Fig. 6d-f. The low-density distribution zone near the origin region of the 0.135 mbar film is more concentrated and uniform over the

entire electric field range, suggesting that the nanodomains in the film increase the polarization switch, resulting in less nonlinear polarization and more stable relaxor ferroelectric states in the 0.135 mbar film, which helps to obtain a slim P - E loop and reduce hysteresis loss [14,41].

Fig. 7a-c shows the unipolar P - E loops of the films grown at different oxygen pressures under an applied electric field of 6 MV·cm⁻¹, and the intersection of the fitted tangent line from the maximum polarization of the P - E loops with the polarization axis is the spontaneous polarization (P_s) [42]. For ferroelectric energy storage film capacitors, the recoverable energy density (W_{rec}) is derived from two components: the non-linear polarization (P_s - P_r , the green color in Fig. 7d) corresponding to ferroelectric domain switching at low electric fields ($P < P_s$) and the linear polarization (P_m - P_s , the red color in Fig. 7d) corresponding to linear dielectric response under high electric fields ($P > P_s$) [42]. It is noted that the film grown at 0.135 mbar has the smallest P_s and its linear polarization contribution is the largest, while it has the smallest non-linear polarization contribution, as shown in Fig. 7d. These characteristics are extremely beneficial for getting higher energy storage efficiency. In addition, we extracted P_m and P_r from the P - E loops of BT-BMH films and calculated P_m/P_r , as shown in Fig. 7e. It can be clearly observed that P_m increased with increasing oxygen pressure. The increased P_m can be attributed to two factors: one is the increased Bi ion content, as the Bi 6p and O 2p electrons exhibit strong orbital hybridization. The other is the decreased defect dipoles, which reduces the polarization pinning. This also induces the ϵ_r increase from 81.3 to 100.3 with increasing oxygen pressure (Fig. 7f). As for P_r , the film grown at 0.135 mbar exhibits the smallest P_r ~ 2.037 μ C·cm⁻², while the 0.170 mbar film has the largest P_r ~ 3.058 μ C·cm⁻². In BT-BMH films, the defect dipoles formed by O vacancies and Bi vacancies produce defect polarizations (P_D) aligned along the spontaneous polarization (P_s) direction, but the defect dipole moments, and associated P_D slowly response to the electric field due to the pinning effect of vacancies [43]. Upon removal

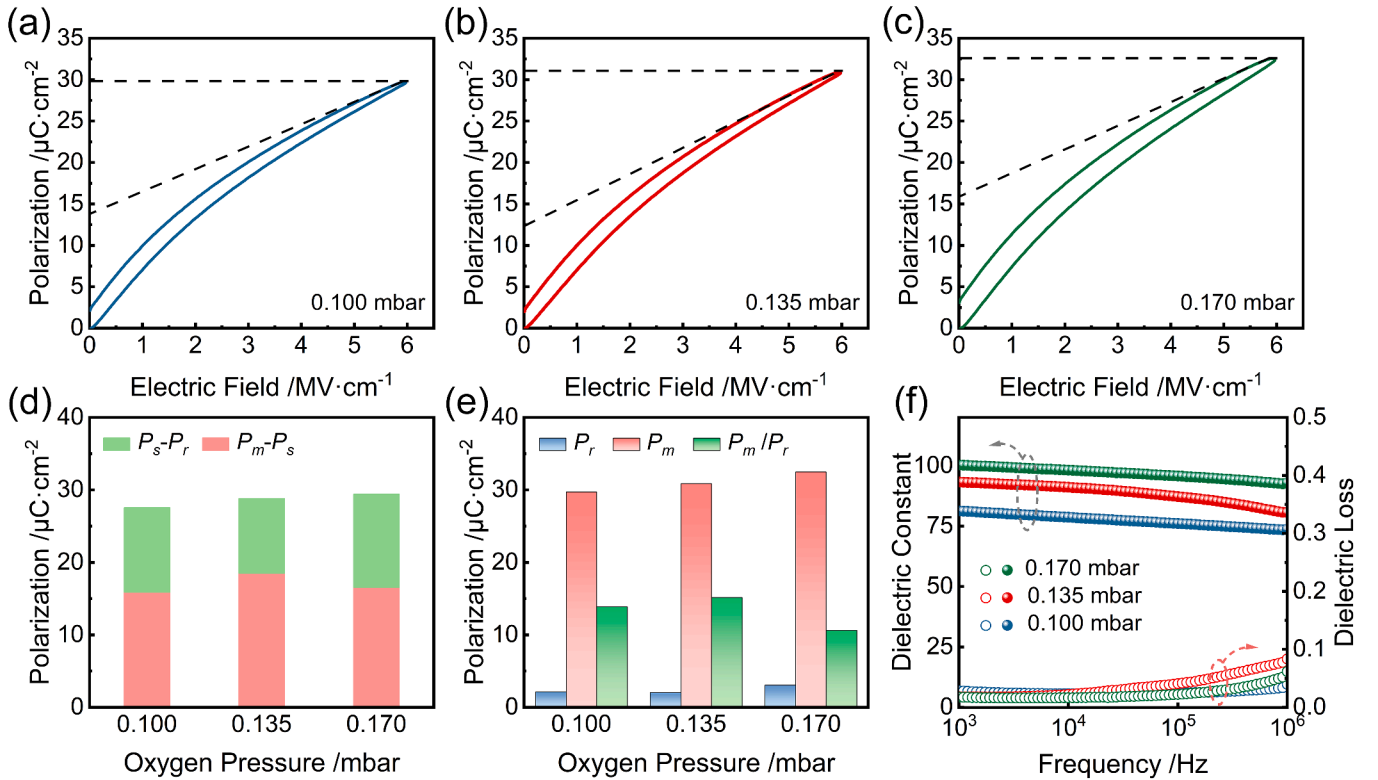


Fig. 7. (a)-(c) Unipolar P - E loops under 6.0 MV·cm⁻¹ for the BT-BMH films under different oxygen pressures. (The intercept of the black line corresponds to spontaneous polarization.) (d) Summary of polarization mechanisms at 6.0 MV·cm⁻¹, red color stands for linear-type polarization and green color for non-linear type polarization. (e) Comparison of P_m , P_r and P_m/P_r of BT-BMH films under an electric field of 6.0 MV·cm⁻¹. (f) Frequency dependence of the dielectric constant (ϵ_r) and dielectric loss ($\tan \delta$) at room temperature for the BT-BMH films.

of the applied electric field, the unswitched oxygen and Bi vacancies, along with the associated P_D will provide an intrinsic restoring force to help the P_s switch to its original orientation, which leads to a small P_r , as shown in Fig. S2. However, when the concentration of defect dipole is above a certain value, it will exacerbate the domain wall pinning and increase P_r [34]. Besides, oxygen vacancies usually pin the movement of the domain walls and induce P_r increase. The excessive oxygen vacancies in the film grown at 0.170 mbar film leads to an increase in P_r . The balanced defect dipoles and oxygen vacancies in the film grown at 0.135 mbar induce the smallest P_r . As a result, the film grown at 0.135 mbar exhibit the highest P_m/P_r .

From the above discussion, we know that the defect dipole and oxygen vacancy significantly affect the breakdown strength and polarization behavior, which definitely influence the energy storage performances. The energy storage density and efficiency as a function of the electric field are derived from the unipolar P - E loops (Fig. 8a-c). It is clearly seen that the energy density of the film grown at 0.135 mbar, which shows the highest breakdown strength and P_m/P_r , achieves to $108.9 \text{ J}\cdot\text{cm}^{-3}$ with η of 79.6 % at room temperature, as shown in Fig. 8d. To better evaluate the application in pulsed power devices, the pulsed power charge and discharge tests were carried out as shown in Fig. 9a. The power density of the film grown at 0.135 mbar can go up to $20.17 \text{ MV}\cdot\text{cm}^{-3}$ with a fasting discharge time of $3.83 \mu\text{s}$. For practical applications, dielectric capacitors are expected to have excellent fatigue stability and thermal stability to ensure long-term stable operation of the equipment under extreme operating conditions. Fig. 9b shows the W_{re} and η of the film grown at 0.135 mbar after 10^6 fatigue cycles under an applied electric field of $6.0 \text{ MV}\cdot\text{cm}^{-1}$ at room temperature. It can be seen

that its energy storage performance did not undergo any significant degradation after 10^6 cycles, with the change rate of the W_{re} and η is only 0.5 % and 2.1 % respectively, demonstrating that the film has strong fatigue resistance properties due to the highly dynamic relaxation nanodomain structure. Besides, the temperature-dependent energy storage performance of BT-BMH films was characterized in the temperature range from -100 to 350°C is shown in Fig. 9c, and its corresponding unipolar P - E loops are shown in Fig. S3. The energy storage density of the film grown at 0.135 mbar is the largest among these three films and can go up to $\sim 69.1 \text{ J}\cdot\text{cm}^{-3}$ with energy storage efficiency of $\sim 73.3\%$, owing to the highest breakdown strength and slim P - E loops. Moreover, the change rate in this temperature range is $<10\%$, which exhibits excellent thermal stability and meets the requirement of industrial applications. The excellent thermal stability of BT-BMH film grown at 0.135 mbar is attributed to its stable dielectric constant and low dielectric loss within the tested temperature range in Fig. S4. As shown in Fig. 9d, the synergistic effect of defect dipole and oxygen vacancy make the BT-BMH film grown at 0.135 mbar exhibit a great advantage in energy storage performance over such wide temperature range, compared with other Pb-based and Pb-free film capacitors [44–52].

4. Conclusions

In summary, we design the defect dipoles and oxygen vacancies in the BT-BMH film to improve its energy storage performance by controlling the oxygen pressure during the film deposition process. It is found that the concentration of oxygen vacancy increases as the oxygen

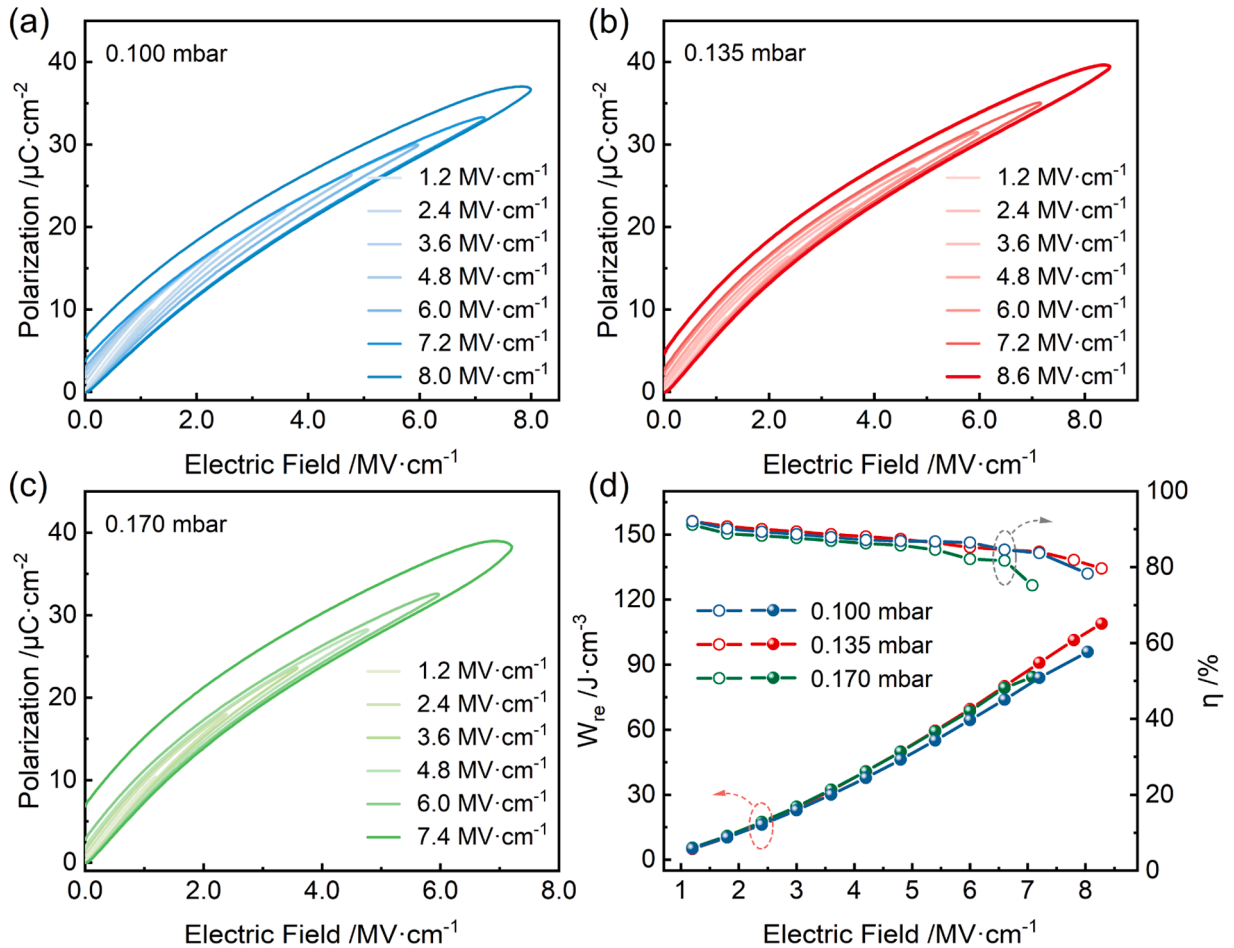


Fig. 8. Unipolar P - E loops of BT-BMH films grown at 0.100 mbar (a), 0.135 mbar (b), 0.170 mbar (c) under different electric fields. (d) Energy density and efficiency of the BT-BMH films as functions of electric field.

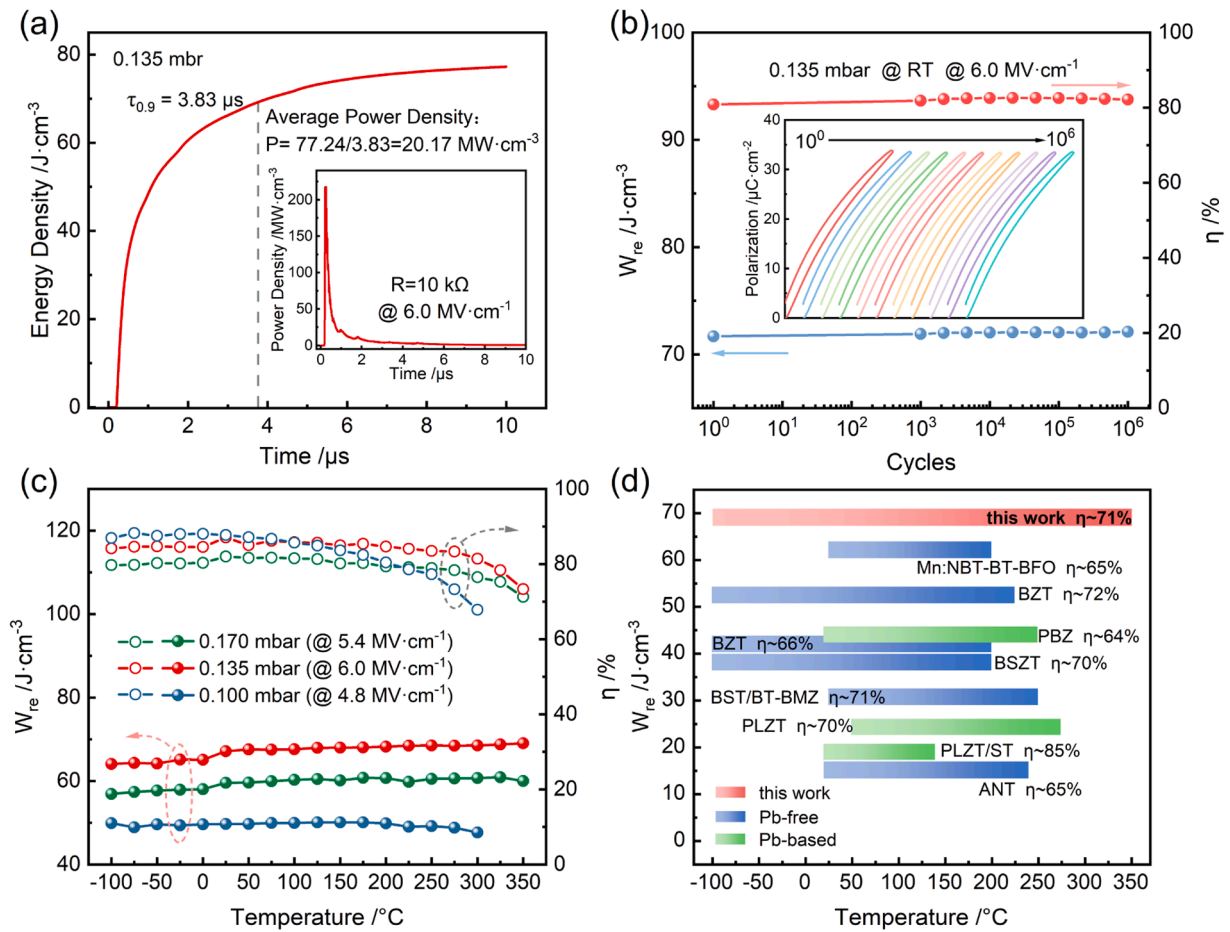


Fig. 9. (a) The discharge energy density as a function of discharge time for the BT-BMH film of 0.135 mbar, the inset is the power density as a function of discharge time. (b) Energy storage performance of BT-BMH film grown at 0.135 mbar under 6.0 MV $\cdot cm^{-1}$ over 10^6 charging-discharging cycles at room temperature; the inset is the corresponding fatigue tested unipolar $P-E$ loops. (c) Temperature dependence of energy storage performances of the BT-BMH films over the temperature range from -100 to 350 $^{\circ}C$. (d) Comparison of energy storage performance of the BT-BMH film grown at 0.135 mbar in this work with representative Pb-based and Pb-free dielectric films.

pressure raise due to the formation of Bi-O defect dipole. On the one hand, the oxygen vacancies act as the trapping center to block the charge carriers transport in the film and increase the breakdown strength, but when the concentration of oxygen vacancies is above a certain value, it will provide a conductive path for charge carriers and make the breakdown strength deteriorate. On the other hand, defect dipoles can interact with free charges to create deep-level trap states, which can effectively suppress carrier activation and hence reduce leakage current, but high concentrations of defect dipoles tend to cause local breakdown of the film. Moreover, the excess oxygen vacancies and defect dipoles will pin the movement of the domain walls and induce P_r increases. The suitable defect dipole can reduce P_r due to it can generate nanodomain and lower the domain switching energy barrier. The balanced defect dipoles and oxygen vacancies in the film grown at 0.135 mbar make its energy storage density go up to 108.9 $J \cdot cm^{-3}$ accompanied by an efficiency of 79.6% at room temperature and excellent thermal stability in the wide temperature range of -100 to 350 $^{\circ}C$ with a high W_{re} of 69.1 $J \cdot cm^{-3}$ is also achieved. Our research demonstrates a paradigm of modulating energy storage properties via defect dipole and oxygen vacancy engineering, and provides a practical and efficient strategy for the design of novel dielectric materials with excellent energy storage performance.

CRediT authorship contribution statement

Weijie Fu: Writing – original draft, Methodology, Formal analysis,

Data curation, Conceptualization. **Yi-qin Lu:** Validation, Investigation. **Qiuyang Han:** Validation, Investigation. **Tian-Yi Hu:** Validation, Investigation. **Tingzhi Duan:** Validation, Investigation. **Yupeng Liu:** Validation, Investigation. **Shao-Dong Cheng:** Investigation. **Yanzhu Dai:** Investigation. **Ming Liu:** Writing – review & editing, Supervision, Methodology, Formal analysis, Conceptualization. **Chunrui Ma:** Writing – review & editing, Supervision, Methodology, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in

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